

Geographic atrophy: Mechanism of disease, pathophysiology, and role of the complement system.

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Geographic atrophy (GA) is an advanced form of age-related macular degeneration (AMD), characterized by atrophic lesions that first start in the outer retina and progressively expand to cover the macula and the fovea, the center of the macula, leading to irreversible loss of vision over time. GA is distinct from wet or neovascular AMD (nAMD), the other form of advanced AMD. Neovascular AMD is characterized by new invading leaky blood vessels in the macula that can lead to acute vision loss. GA and nAMD may coexist in the same eye. The underlying pathophysiology of GA is complex and thought to involve chronic inflammation due to overactivation of the complement system that leads to the loss of photoreceptors, retinal pigment epithelium (RPE), and the underlying choriocapillaris. The disappearance of these structures appears as sharply demarcated atrophic lesions that are typical of GA. Researchers have reported about 1 million reported cases of GA in the United States, and about 160,000 cases occur per year. The most important risk factors for GA are increasing age and family history. Diagnosis of GA is usually made by using multimodal imaging techniques. Lesions associated with GA are highly heterogeneous, and the growth rate may differ from patient to patient. Despite the progressive nature of GA, the fovea may be spared until much later in the disease, thereby retaining central vision in patients. With time, atrophic lesions may progressively grow to involve the fovea, thereby severely impairing central vision. Vision loss can happen rapidly once the lesions reach the fovea. However, even without the involvement of the fovea, ongoing vision impairment impacting daily life may be present. Median time from GA not involving the center of the fovea (without subfoveal involvement) to GA with lesion boundary affecting the foveal center (subfoveal involvement) ranges from 1.4 to 2.5 years. GA can greatly impact patients' functioning and quality of life and limit their independence by interfering with activities of daily living, including difficulties with reading, driving, watching television, recognizing faces, and being unable to do household chores. No treatments have been available until intravitreal pegcetacoplan was recently approved by the US Food and Drug Administration for GA secondary to AMD.

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Characterizing Patient Perceptions of Living with Geographic Atrophy: The Global Geographic Atrophy Insights Survey.

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Geographic atrophy (GA) is an advanced form of age-related macular degeneration leading to irreversible vision loss and negative impacts on quality of life. To assess the experiences of living with GA, the Geographic Atrophy Insights Survey (GAINS) was conducted between October 12, 2021, and December 10, 2021, captured the responses of individuals ≥ 60 years with a self-reported GA diagnosis residing in

the United States, Canada, Australia, and six European countries. Survey questions focused on the perceptions of individuals living with GA and covered six themes: speed of disease progression, effect on independence, impact on quality of life, emotional toll of GA, misconceptions and need for further education about GA, and clinician interactions. An exploratory comparison between participants with unilateral and bilateral GA was conducted. The survey included 203 individuals with a mean age of 70 years; 42% had bilateral GA. Most respondents (77%) agreed ("strongly" or "somewhat agreed") that GA impacted their vision faster than expected, and 68% agreed that it is hard to enjoy life fully the way they did before GA diagnosis. Regarding comparisons between individuals with bilateral and unilateral GA, both groups reported similar "major" or "moderate" negative impacts on their ability to drive (73% vs 75%, respectively), followed by the ability to read (66% vs 71%), and ability to travel as much as they would prefer (62% vs 62%). Among participants, 49% and 56% of respondents with bilateral and unilateral GA, respectively, reported major/moderate negative impacts on self-confidence and 40% of both cohorts reported major/moderate negative impacts on mental health. Our survey provides further insight on the burden experienced by individuals living with GA. We find similar responses between unilateral and bilateral GA groups, highlighting the impact GA may have on an individual's quality of life even when only one eye is affected.

Structural Characteristics of Retinal Layers Adjacent to Geographic Atrophy.

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To compare the anatomy of different retinal layers adjacent to areas of geographic atrophy (GA) to those of eyes with no known ocular diseases. Spectral-domain optical coherence tomography (SD-OCT) scans from eyes with GA were retrospectively reviewed. Two scans with no findings suggestive of GA changes on OCT were selected from immediately above and/or below the edge of the lesions. Thickness values of the retinal layers were calculated and compared to values obtained from normal subjects. Forty-four eyes (30 patients) were compared to 20 healthy eyes. Retinal pigment epithelium (RPE), inner nuclear layer (INL), and full retinal thickness (FRT) values were significantly lower in patients compared to healthy subjects. Thicknesses of all other layers were not significantly different. Clinically appearing, non-involved RPE and INL layers of eyes with GA demonstrate significant thinning compared to corresponding layers in eyes with no known ocular diseases.

Geographic atrophy: pathophysiology and current therapeutic strategies.

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Geographic atrophy (GA) is an advanced stage of age-related macular degeneration (AMD) that leads to gradual and permanent vision loss. GA is characterized by the loss of photoreceptor cells and retinal pigment epithelium (RPE), leading to distinct atrophic patches in the macula, which tends to increase with time. Patients with geographic atrophy often experience a gradual and painless loss of central vision, resulting in difficulty reading, recognizing faces, or performing activities that require detailed vision. The primary risk factor for the development of geographic atrophy is advanced age; however, other risk factors, such as family history, smoking, and certain genetic variations, are also associated with AMD. Diagnosis is usually based on a comprehensive eye examination, including imaging tests such as fundus photography, optical coherence tomography (OCT), and fluorescein angiography. Numerous clinical trials are underway, targeting identified molecular pathways associated with GA that are

promising. Recent approvals of Syfovre and Izervay by the FDA for the treatment of GA provide hope to affected patients. Administration of these drugs resulted in slowing the rate of progression of the disease. Though these products provide treatment benefits to the patients, they do not offer a cure for geographic atrophy and are limited in efficacy. Considering these safety concerns and limited treatment benefits, there is still a significant need for therapeutics with improved efficacy, safety profiles, and better patient compliance. This comprehensive review discusses pathophysiology, currently approved products, their limitations, and potential future treatment strategies for GA.

Gene therapy for geographic atrophy in age-related macular degeneration: current insights.

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Geographic atrophy (GA) is the advanced stage of non-neovascular (dry) age-related macular degeneration, defined by the presence of sharply demarcated atrophic lesions of the outer retina. The complement system is integral to the body's natural immune response, and hence its overactivation can lead to tissue damage and inflammation. It has been shown to play a significant role in GA lesion development and progression, and therefore, complement inhibition is emerging as a promising avenue for therapeutic intervention. With the recent approval by the Food and Drug Administration of drugs like SYFOVRE™ (pegcetacoplan injection) and IZERVAY™ (avacincaptad pegol intravitreal solution), there is hope for the development of interventions capable of slowing down or arresting the progression of GA. In particular, gene therapy intervention is gaining traction for halting GA atrophy at the source of our genes. The concept is to insert a gene into the eye that will act as an ocular "bio-factory," producing a desired protein. This can either lead to overproduction of an already available protein or produce a substance not typically generated in the eye. This review aims to provide an overview of the present understanding of GA, encompassing risk factors, prevalence, pathophysiology, and genetic associations. It will also highlight the current landscape of GA treatment, with particular emphasis on gene therapy intervention.

Treating patients with geographic atrophy: are we there yet?

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Geographic atrophy (GA) is a progressive degenerative disease that significantly contributes to visual impairment in individuals aged 50 years and older. The development of GA is influenced by various modifiable and non-modifiable risk factors, including age, smoking, and specific genetic variants, particularly those related to the complement system regulators. Given the multifactorial and complex nature of GA, several treatment approaches have been explored, such as complement inhibition, gene therapy, and cell therapy. The recent approval by the Food and Drug Administration of pegcetacoplan, a complement C3 inhibitor, marks a significant breakthrough as the first approved treatment for GA. Furthermore, numerous interventions are currently in phase II or III trials, alongside this groundbreaking development. In light of these advancements, this review provides a comprehensive overview of GA, encompassing risk factors, prevalence, genetic associations, and imaging characteristics. Additionally, it delves into the current landscape of GA treatment, emphasizing the latest progress and future considerations. The goal of starting this discussion is to ultimately identify the most suitable candidates for each therapy, highlight the importance of tailoring treatments to individual cases, and continue

monitoring the long-term implications of these emerging interventions.

Genetic and molecular biomarkers for geographic atrophy.

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Geographic atrophy (GA) is characterized by atrophy of the retina, retinal pigment epithelium and choriocapillaris, causing a gradual loss of vision over time. Treatment options to prevent initiation or progression of GA are limited; two recently FDA-approved inhibitors of the complement system (pegcetacoplan, avacincaptad pegol) showed a modest decrease in GA lesion growth in phase 3 clinical trials. Exploration of genetic and molecular biomarkers in GA plays a critical role in our battle against this blinding disease to improve early disease detection, to find more effective therapies, and to provide personalized care to patients. In this review, we provide a comprehensive overview of the current literature investigating genetic and molecular biomarkers for GA. Genetic studies identified multiple genes and variants that play a role in progression to GA and GA lesion growth, involving pathways such as complement activation, extracellular matrix interaction and lipid metabolism. The number of published studies assessing molecular biomarkers for GA initiation and progression in ocular matrices is limited. Several studies evaluated molecular biomarkers in the systemic circulation, showing higher levels of complement activation and a causal role of lipid subfractions in GA. Larger, well-powered studies are needed to identify novel and validate existing biomarkers, and to investigate the potential of combining genetic and molecular markers with imaging techniques for more accurate diagnosis and monitoring of GA. The development of personalized medicine approaches based on individual genetic and molecular profiles could hold promise for more effective and targeted treatments for this devastating disease.

The Progression of Geographic Atrophy Secondary to Age-Related Macular Degeneration.

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Geographic atrophy (GA) is an advanced form of age-related macular degeneration (AMD) that leads to progressive and irreversible loss of visual function. Geographic atrophy is defined by the presence of sharply demarcated atrophic lesions of the outer retina, resulting from loss of photoreceptors, retinal pigment epithelium (RPE), and underlying choriocapillaris. These lesions typically appear first in the perifoveal macula, initially sparing the foveal center, and over time often expand and coalesce to include the fovea. Although the kinetics of GA progression are highly variable among individual patients, a growing body of evidence suggests that specific characteristics may be important in predicting disease progression and outcomes. This review synthesizes current understanding of GA progression in AMD and the factors known or postulated to be relevant to GA lesion enlargement, including both affected and fellow eye characteristics. In addition, the roles of genetic, environmental, and demographic factors in GA lesion enlargement are discussed. Overall, GA progression rates reported in the literature for total study populations range from 0.53 to 2.6 mm²/year (median, ~1.78 mm²/year), assessed primarily by color fundus photography or fundus autofluorescence (FAF) imaging. Several factors that could inform an individual's disease prognosis have been replicated in multiple cohorts: baseline lesion size, lesion location, multifocality, FAF patterns, and fellow eye status. Because best-corrected visual acuity does not correspond directly to GA lesion enlargement due to possible foveal sparing, alternative assessments are

being explored to capture the relationship between anatomic progression and visual function decline, including microperimetry, low-luminance visual acuity, reading speed assessments, and patient-reported outcomes. Understanding GA progression and its individual variability is critical in the design of clinical studies, in the interpretation and application of clinical trial results, and for counseling patients on how disease progression may affect their individual prognosis.

Measuring geographic atrophy in advanced age-related macular degeneration.

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To present a method developed for measuring areas of geographic atrophy (GA) in advanced age-related macular degeneration, A microfilm reader projected the 30 degrees fundus photograph of the macula. Retinal landmarks, atrophic areas, and spared areas within the atrophy were traced, without access to drawings of other years. The total atrophic area was calculated, as was the atrophy within a four-disc-area circle entered on the estimated foveal center. The configuration of the atrophy was documented. Avoidable sources of discrepancy included variability in peripapillary atrophy seen on the photograph, and variability seen in the extent of the field. Reproducibility studies found a median absolute difference of 0.19 Macular Photocoagulation Study disc areas (DA) in total atrophy between repeat drawings, with 75% of repeat drawings having a difference of less than 0.33 DA. For central atrophy measures, there was a median difference of 0.08 DA, with 75% of pairs having a difference of less than 0.18 DA. Features making the definition of borders of GA difficult include the presence of drusen and pigmentary alteration, a fundus in which choroidal vessels are easily visible, and variation in the appearance of GA within a single area of atrophy. This method provides a reliable means of measuring the size of atrophic areas in GA and will be useful for measuring longitudinal change. It may be difficult to determine whether central spared areas are present, and correlation with visual acuity and macular perimetry may be helpful.

Evolution of geographic atrophy of the retinal pigment epithelium.

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The aim of this study was to trace the evolution of geographic atrophy (GA) by clinical documentation and by clinico-morphological correlation in representative eyes. Geographic atrophy commonly commenced within a parafoveal band of incipient atrophy of varying width, characterised by semisolid drusen and a microreticular pigment pattern. Progression of atrophy mostly skirted fixation and visual acuity was a poor guide to the functional impact, an estimate of the percentage of fovea involved proving a more useful clinical parameter. The rate of progression slowed once GA had involved all the retina affected by incipient atrophy and the risk of choroidal neovascularization appeared to decline. An earlier histological classification of the evolution of GA is revised according to the ultrastructural findings. Membranous debris was not previously recognised and its contribution to the findings in incipient atrophy and to dot-like drusen is described.